

Niramay M. Patel

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Education

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- Indian Institute of Science Education and Research (IISER), Bhopal** *August 2023 – Present*
BS in Engineering Science (Major in Computer Science)
- CPI: 9.50/10.0
 - **Completed Courses:** Linear Algebra, Calculus, Multi-variable Calculus, Discrete Mathematics, Data Structure and Algorithm, Probability and Statistics, Signals and System.
 - **Ongoing Courses:** Machine Learning, Computer Vision, Theory of Computation, Fundamentals of Database Systems.
- Atmiya Vidya Mandir, Surat, Gujarat** *Passed in March 2023*
 ◦ CBSE Std. 12 Result: 94.8%

Experience

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- Research Intern (Remote)** *IBM and IIT Roorkee*
IBM Research in collaboration with IIT Roorkee *August 2025 – Present*
- Working on advancing **speech-centric large language models (LLMs)** for multilingual and low-resource settings.
 - Exploring methods that integrate external knowledge and retrieval mechanisms to improve robustness and translation quality.
- Research Intern (On site)** *VisDom Lab, IISERB*
Professor: Dr. Vinod Kurmi *March 2025 – Present*
- Worked on **fairness in deep learning**, analyzing SEBRA under label noise, studying bias propagation in **ParlerTTS**, and exploring coreset selection for fairness-aware training.
 - This work led to a first-author paper on regularization for robust generalization under distribution shifts.
- Research Intern (Remote)** *IIT Roorkee, India*
Prof: Dr. Raksha Sharma *Jan 2025 – July 2025*
- Conducted a research project focusing on improving the robustness of **speech-to-text models** under real-world noise conditions.
 - Designed a structured **hyperparameter tuning framework** and proposed a novel evaluation protocol to quantify performance stability across diverse acoustic settings.
- Research Intern (On site)** *MOON Lab, IISERB*
Prof: Dr. P. B. Sujit *Dec 2024 – Jan 2025*
- Explored core topics in **Computer Vision**, including OpenVINS, MSKCF, SLAM algorithms, and the mathematical foundations of sensor fusion.
 - Gained strong theoretical and practical expertise by implementing **Visual Odometry** on the KITTI dataset.
- Project Intern** *Remote*
Personifywy *March 2024 – May 2024*
- Built machine learning classifiers including a **Hate Speech Detector** (Decision Tree, 88% accuracy) and a **Spam SMS Classifier** (Naive Bayes) to improve online safety and reduce unsolicited messages.
 - Performed **credit exploratory data analysis (EDA)** to identify key factors influencing creditworthiness from raw financial datasets.

Publications

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- **Regularization for Robust Generalization under Distribution Shifts** Under review at ICLR 2026
 - **Object-Centric Attention for Bias-Robust Vision** Under review at WACV 2026
 - **Hyperparameter Strategies for Noise-Robust Speech-to-Text** Under review at AACL 2025

Projects

Visual Odometry

[github repo](#) 

MOON LABS, December 2024

- Estimated the camera's motion from a sequence of images, a crucial task in autonomous driving and SLAM. I successfully reconstructed the **vehicle's trajectory** and evaluated the accuracy using ground truth data.
- Tools used: Python, OpenCV, Physics, Feature extraction and matching.

Bollywood Celebrity Face Matching

[github repo](#) 

Personal Project, December 2024

- A **transfer learning** project utilizing pre-trained ResNet50 CNN to match an uploaded face with the most likely Bollywood celebrity.
- Tools used: Python, CNN, Deep Learning, Streamlit, Scikit-learn.

Credit EDA

[github repo](#) 

Personify Internship, October 2024

- Performed EDA on data sets with more than **300,000 rows and 122 columns**. It involves exhaustive data preprocessing, visualization and analysis. Finally, a Logistic Regression model is trained to verify the effectiveness of the data preprocessing steps.
- Tools used: Jupyter Notebook, NumPy, Pandas, Matplotlib, Scikit-learn

Personal Website

[github repo](#) 

Personal Project, June 2024

- Developed a personal portfolio website and hosted it on Render for seamless deployment and accessibility.
- Added interactive elements and dynamic design to make it appealing.
- Tools used: HTML, CSS, JavaScript, GitHub

Certifications

Machine Learning Specialization by Stanford University and DeepLearning.AI

[certificate](#) 

- Topics covered:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders, and Reinforcement Learning.

Co-Curricular and Extra-Curricular

- Proposed an **AI based Dynamic Bus route rationalization** model that would be required based on real-time traffic and road parameters for the **Delhi Government** in Smart India Hackathon.
- Secured a rank **under 500** as a College Ambassador for **Techfest, IIT Bombay**.
- Attended seminar on **HyWay: Hybrid Hallway** by Dr.Venkat Padmanabhan, Managing Director at **Microsoft Research India**.
- Participated in panel discussion during **Industry-Academia conclave 2025** at IISER Bhopal.
- Developed a dynamic frontend for a **bus travel management website** for the institute.
- Led the **core committee** in organizing a two-day Navaratri event attended by over 1,000 participants, managing end-to-end planning and execution.
- I enjoy playing table tennis, badminton, and cricket to relax during my free time.